



Spectrum-based deep learning framework for dermatological pigment analysis and simulation

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ABSTRACT

Background: Deep learning in dermatology presents promising tools for automated diagnosis but faces challenges, including labor-intensive ground truth preparation and a primary focus on visually identifiable features. Spectrum-based approaches offer professional-level information like pigment distribution maps, but encounter practical limitations such as complex system requirements.

Methods: This study introduces a spectrum-based framework for training a deep learning model to generate melanin and hemoglobin distribution maps from skin images. This approach eliminates the need for manually prepared ground truth by synthesizing output maps into skin images for regression analysis. The framework is applied to acquire spectral data, create pigment distribution maps, and simulate pigment variations.

Results: Our model generated reflectance spectra and spectral images that accurately reflect pigment absorption properties, outperforming spectral upsampling methods. It produced pigment distribution maps with correlation coefficients of 0.913 for melanin and 0.941 for hemoglobin compared to the VISIA system. Additionally, the model's simulated images of pigment variations exhibited a proportional correlation with adjustments made to pigment levels. These evaluations are based on pigment absorption properties, the Individual Typology Angle (ITA), and pigment indices.

Conclusion: The model produces pigment distribution maps comparable to those from specialized clinical equipment and simulated images with numerically adjusted pigment variations. This approach demonstrates significant promise for developing professional-level diagnostic tools for future clinical applications.

1. Introduction

The application of deep learning in dermatology has gained increasing attention as a potential tool for automated diagnosis, with numerous studies demonstrating its efficacy in classifying and segmenting skin diseases [1,2]. However, despite success in automating feature extraction, deep learning approaches encounter practical limitations, such as the labor-intensive creation of ground truths for training datasets, which becomes more challenging and error-prone as image volumes expand. Additionally, the demand for high-resolution images in clinics is constrained by computational resources, with current studies mainly focusing on the automatic classification and segmentation of skin conditions easily identifiable by non-specialists. This necessitates methodologies that offer deeper clinical insights, such as pigment distribution maps, typically only accessible through specialized equipment like the VISIA system [3].

Reflectance spectra reveal molecular-level differences [4], often invisible to the naked eye, driving extensive research in dermatology into applications ranging from pigmentation [5] and hydration [6]

to acne [7], skin cancer [8], and vein visualization [9]. Our study specifically focuses on skin pigmentation, exploiting the differential light absorption at various wavelengths. While point measurements using a spectroscope provide high-resolution spectral data, they are limited in depicting the spatial heterogeneity of skin characteristics. Therefore, our emphasis shifts to spectral imaging for a comprehensive analysis of pigment distribution.

Spectral imaging, which combines spectroscopy and imaging, is categorized into multi-spectral and hyper-spectral imaging based on the number of spectral bands captured [10]. Despite offering detailed pixel-wise spectral information, it confronts limitations that obstruct its broader application, such as the need for additional optical components [11], elevated costs, and the risk of motion artifacts during lengthy captures. Chromatic aberration further complicates high-quality image acquisition. Alternatives like camera sensors with integrated spectral filters necessitate trade-offs between image resolution and wavelength coverage. Addressing these challenges, our study investigates spectral upsampling methods from computer graphics to

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reconstruct reflectance spectra from color images, thereby avoiding the complexities of direct spectral imaging. This approach has proven effective in basic applications, such as with color checker patterns [12–14]. Our research expands this concept to skin pigment analysis, employing the unique optical properties of melanin and hemoglobin to extract detailed pigment information from skin images.

This study introduces a spectrum-based deep learning framework for the analysis and simulation of skin pigmentation. The framework trains a model to generate melanin and hemoglobin distribution maps from skin images, utilizing the optical properties of these pigments to derive skin reflectance spectra and synthesize images that resemble camera captures. We validate our methodology's effectiveness by comparing our skin reflectance spectra with those from spectral upsampling methods, assessing our pigment distribution maps against those from the VISIA system, and by creating simulations of pigment variation. These evaluations are based on pigment absorption properties, the Individual Typology Angle (ITA), and pigment indices.

2. Methods

The frameworks for training a deep learning model and its application, as detailed in Fig. 1, merge biophysical principles from computer vision [15] with dermatology, validated through preliminary research [16–18]. While prior studies using a spectrum-based approach primarily focused on the application's performance, they lacked intermediate processes such as the generation of reflectance spectra and spectral images, as well as spectral interpretation and verification of application results. Additionally, the integration and structuring of the overall training and application frameworks were not sufficiently developed. This study, therefore, aims to further explore spectral interpretations based on the optical properties of skin pigments and to compare these generated reflectance spectra with those from existing spectral upsampling methods. Moreover, it introduces comprehensive frameworks for both the training and application phases, embedding dermatology research validation techniques. The subsequent sections will discuss the details of each framework.

2.1. Training framework

The component separation model in this study is trained on synthesized images derived from pigment maps. While this method may extend training time and pose challenges in matching the performance of conventional regression analysis methods that utilize actual ground truths, it offers distinct advantages. Primarily, this approach eliminates the need for the manual preparation of ground truth data. Furthermore, the framework's ability to extract pigment maps from skin images and subsequently synthesize skin images from these maps significantly enhances its utility, allowing for numerically controlled simulation of pigment variation.

The study utilizes cross-polarized images as input, leveraging their widespread use in pigment analysis [19,20]. This imaging technique improves the visibility of pigment regions and minimizes specular reflections by capturing diffused light that reflects from the skin's deeper layers. For analyses that also need to consider skin surface morphology, such as wrinkles and pores, a transition to standard color images would be required.

2.1.1. Preprocessing

Preprocessing comprises three stages. The initial stage involves distinguishing skin from non-skin regions due to their distinct optical properties. This is achieved using a pre-trained skin segmentation model based on the ResUNet++ architecture [21], with details provided in Section 2.1.4. Trained with 2900 images, including both cross-polarized and standard skin images, the model attained a complete intersection over union (CIoU) [22] of 0.963. After segmentation, the input image is overlaid with a binary mask and divided into

multiple patches. To prevent any visible boundary lines in the assembly of full-sized output maps or skin images, adjacent patches overlap by 15 pixels during division. The final preprocessing step applies inverse gamma correction to linearize the image's color characteristics, converting the color space from standard RGB (sRGB) to linear RGB. This adjustment ensures more accurate color representation and analysis.

2.1.2. Reflectance analysis

In this stage, each patch is processed sequentially through the component separation model, producing melanin, hemoglobin, and shading maps of 256×256 pixels each. The shading map accounts for effects from non-spectral components like shadows due to light direction and is integral to the image synthesis process (Section 2.1.3).

Light absorption within the skin primarily occurs through melanin in the epidermis and hemoglobin in the dermis, with significant scattering by elastin and collagen fibers in the dermis layer [23]. The absorption and scattering properties of the skin layers are understanding skin's interaction with light and are mathematically formulated as follows [24]:

$$\mu_a(\lambda) = \sum_i V_i \epsilon_i(\lambda) C_i \quad (1)$$

$$\mu'_s(\lambda) = a \left(f_{Ray} \left(\frac{\lambda}{\lambda_0} \right)^{-4} + (1 - f_{Ray}) \left(\frac{\lambda}{\lambda_0} \right)^{-b_{Mie}} \right) \quad (2)$$

Here, $\mu_a(\lambda)$ represents the skin's absorption coefficient at wavelength λ , with V_i , ϵ_i , and C_i denoting the volume fraction, extinction coefficient, and concentration of chromophores (primarily melanin and hemoglobin), respectively. $\mu'_s(\lambda)$ is the reduced scattering coefficient, incorporating both Rayleigh and Mie scattering. a , b_{Mie} , and f_{Ray} represent the scattering amplitude, Mie scattering power, and fraction of Rayleigh scattering events, respectively. The wavelength λ ranges from 450 to 750 nm (at intervals of 10 nm), covering the visible spectrum.

Skin reflectance spectra are derived using methods such as Monte Carlo simulation [25], Diffuse Reflectance Spectroscopy [24], and the Kubelka–Munk theory [26]. The simplified form of the Kubelka–Munk model, assuming most light penetrates the epidermis and reflects off the dermis, is expressed as:

$$R(\lambda) = T_{epidermis}(\lambda)^2 R_{dermis}(\lambda) \quad (3)$$

Here, $R(\lambda)$ is the skin reflectance, with $T_{epidermis}(\lambda)$ and $R_{dermis}(\lambda)$ representing the light transmission through the epidermis and reflection from the dermis, respectively. This model facilitates the calculation of skin reflectance for melanin volume fractions from 1.3% to 43% and hemoglobin from 2% to 7% [27,28], along with their absorption spectra [29]. Reflectance data validated in previous studies [15,30] are utilized in this study.

2.1.3. Image synthesis

Most light illuminating the skin surface refracts into the skin layers, with the remainder being specularly reflected. Approximately 4% to 7% of the incident light is specularly reflected, a proportion consistent across different skin colors within the visible spectrum [31]. The refracted light, upon interacting with the tissue, undergoes spectral changes and is then diffusely reflected back into the air before being detected by the camera sensor. In pigment analysis, polarizers filter out specularly reflected light, which is unaffected by pigments, thereby producing cross-polarized images [19,20]. The spectral representation of this process is formulated as:

$$I_m = \int_{450 \text{ nm}}^{750 \text{ nm}} S(\lambda) \cdot R(\lambda) \cdot M_s \cdot Q_m(\lambda) d\lambda \quad (4)$$

Here, I_m is the image detected for each camera channel m (r , g , b). $S(\lambda)$ is the Spectral Power Distribution (SPD) of the light source, $R(\lambda)$ denotes the skin reflectance, M_s is the shading map, and $Q_m(\lambda)$ is the quantum efficiency of the camera sensor. These elements are

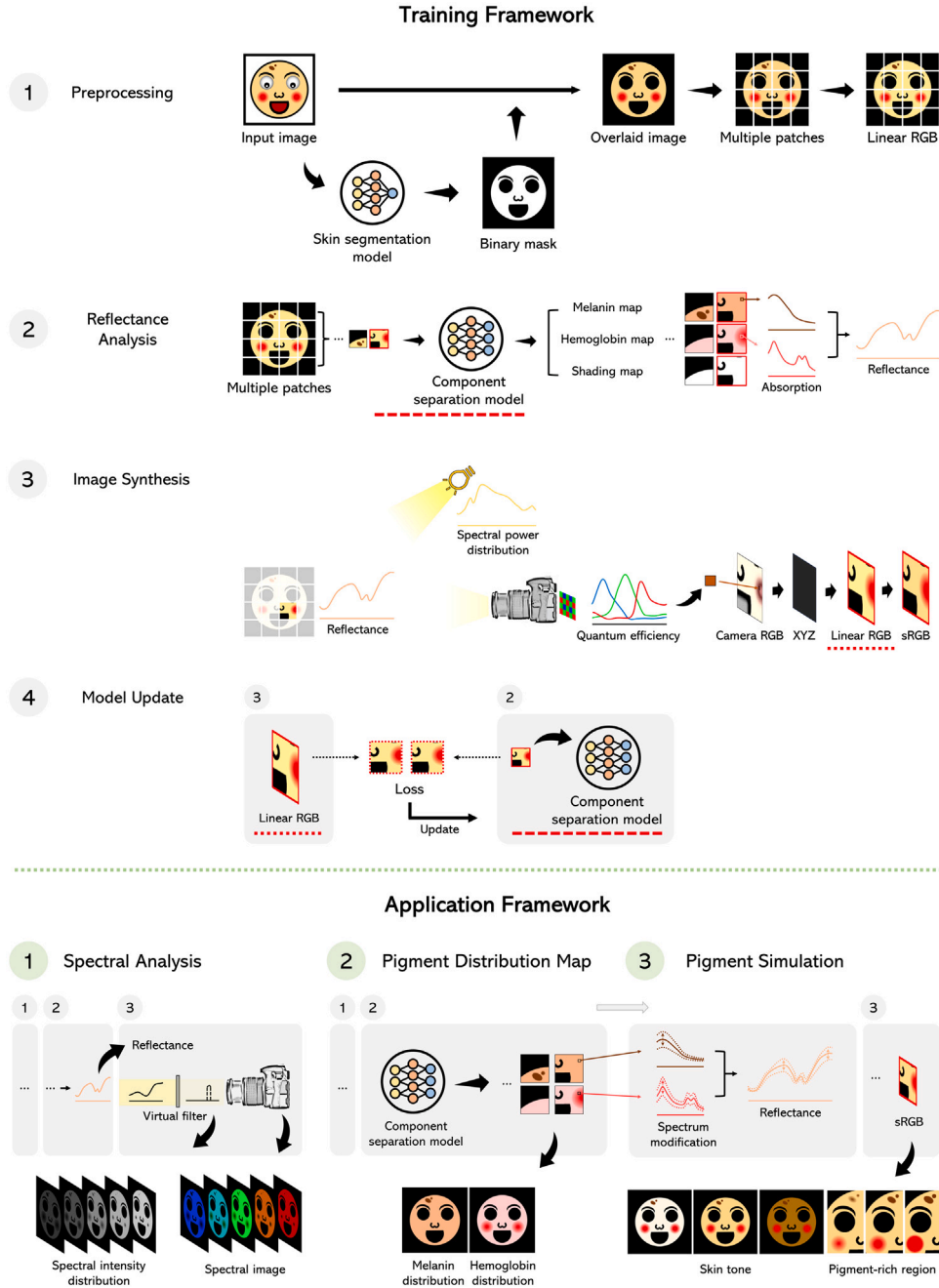


Fig. 1. Overview of the Training and Application Frameworks. The training framework includes four main steps: (1) Preprocessing, involving skin segmentation and patch division for efficient high-resolution image analysis. (2) Reflectance Analysis, to generate melanin, hemoglobin, and shading maps for constructing theoretical skin reflectance spectra. (3) Image Synthesis, combining reflectance, illumination, and camera sensor spectra to create synthesized images resembling camera outputs. (4) Model Update, an iterative process refining the component separation model to minimize discrepancies between synthesized and actual images. The application framework comprises: (1) Spectral Analysis, using virtual filtering to extract spectral data for comparison with outcomes from spectral upsampling methods. (2) Pigment Distribution Map, integrating patch-sized outputs to create comprehensive maps for diagnostic assessment. (3) Pigment Simulation, adjusting melanin and hemoglobin levels to simulate varying skin tones and pigment concentrations.

multiplied and integrated over the visible wavelength range to create the image.

The captured image, represented as $[I_r, I_g, I_b]$, is normalized by multiplying it with the reciprocal of the light source color detected by the camera sensor, achieving white balance correction. This corrected image is then sequentially converted to the XYZ and RGB color spaces through transformation matrices. For model training, the image is used in its linear RGB format, while gamma correction is applied to convert it into the non-linear sRGB color space for final display.

2.1.4. Model update

The model update process iteratively refines the component separation model to minimize discrepancies between the input and synthesized images, both of which are patch-sized and in the linear RGB color space. The objective is to generate output maps that closely replicate the actual pigment conditions of the skin image. During the training phase, the Mean Squared Error (MSE) loss function is selected for its computational simplicity and ease of calculating derivatives. For validation, the Peak Signal-to-Noise Ratio (PSNR) is used to

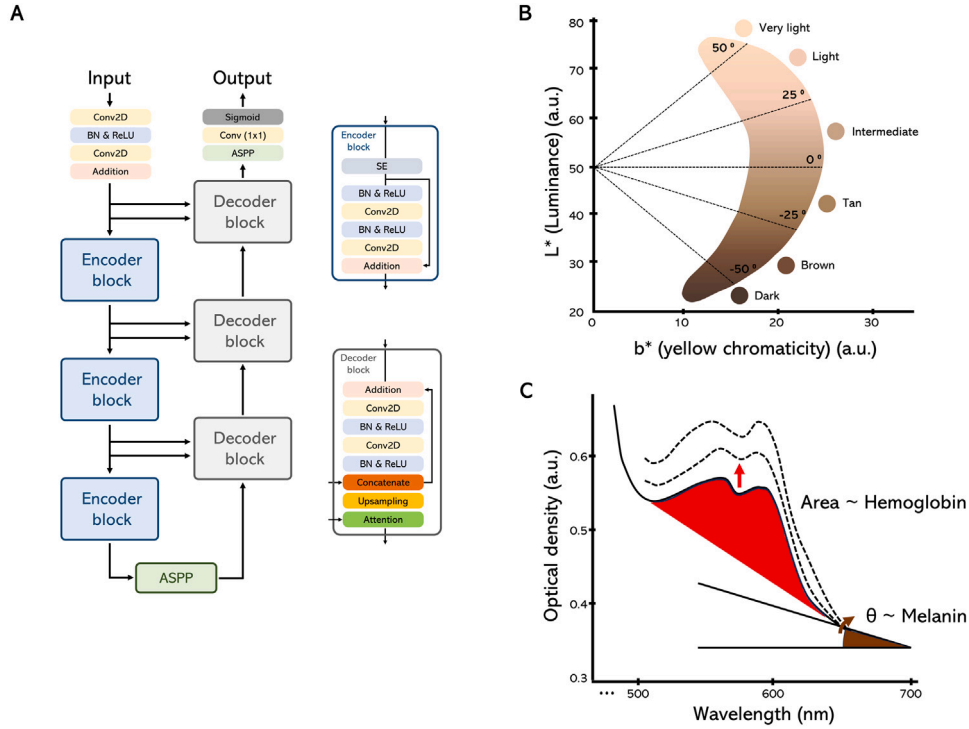


Fig. 2. (A) The ResUNet++ architecture, featuring a stem block, three encoder blocks, atrous spatial pyramid pooling (ASPP) in the bottleneck, three decoder blocks, and a head block. This model incorporates advanced features such as squeeze and excitation blocks (SE), double convolution (two repetitions of batch normalization and ReLU - BN & ReLU), convolution (Conv2D), ASPP, and an attention block. (B) An overview of the Individual Typology Angle (ITA) used for skin tone classification, measured by the angle formed between the L* and a* axes in the Lab* color space. This metric aids in the precise classification of skin tones for enhanced pigment analysis. (C) The optical density spectrum, demonstrating the physiological principles underlying melanin and erythema indices. The angle of optical density (620 to 720 nm) is proportional to melanin content, while the area under the curve (560 to 650 nm) varies with hemoglobin content [23].

better align the model's performance with human visual perception standards.

The architecture of both the component separation model and the skin segmentation model is based on a modified version of ResUNet++ (Fig. 2(A)). This design integrates the ResNet framework with the UNet structure, including a stem block, three encoder blocks (each featuring a residual block and a squeeze & excitation block), a bottleneck with atrous spatial pyramid pooling, three decoder blocks (each with an attention block and a residual block), and a head block. Previous research has demonstrated the effectiveness of ResUNet++ over the standalone ResNet and UNet models [21]. For this study, a lighter version is employed, reducing the number of channels by half to improve computational efficiency.

2.2. Application framework

This section details how the training framework is adapted for dermatological analysis and research applications. Initially, the trained model is deployed to extract spectral reflectance and spectral images, focusing on trends in pigment level variations. This step is fundamental for the development of skin pigment applications, allowing for a direct comparison of outcomes with those derived from three different spectral upsampling methods. Subsequently, comprehensive pigment distribution maps for the entire image are generated by integrating outputs from individual patches. The model's effectiveness in pigment analysis is further validated by comparing it against the VISIA system, specialized clinical equipment. Moreover, by adjusting melanin and hemoglobin concentrations, the framework can simulate various skin tones and pigment-rich regions by varying skin reflectance spectra. These application outcomes are analyzed based on pigment absorption properties, the Individual Typology Angle (ITA), and pigment indices, demonstrating the model's capability for detailed skin analysis.

2.2.1. Spectral analysis

Spectral data acquired include skin reflectance spectra and spectral images. Reflectance spectra for all pixels, derived from the reflectance analysis phase, are multiplied by the light source's SPD during image synthesis. To simulate the effect of an optical band-pass filter, all wavelengths except the target are nullified. This operation, applied across all pixels, yields the spectral intensity distribution for the targeted wavelength. The resulting distribution is further processed by combining it with the camera sensor's quantum efficiency to produce spectral images.

For visual contrast analysis between pigment-rich regions and surrounding normal skin, spectral images are converted to grayscale. These images show varying intensities across different wavelengths, attributed to the wavelength-dependent characteristics of both the light source's SPD and the camera sensor's quantum efficiency. To standardize these images to a consistent average intensity, a post-processing step is implemented, expressed as:

$$I'_{wl} = I_{wl} \cdot \frac{k_{avg}}{\frac{1}{nm} \sum_{i=1}^n \sum_{j=1}^m I_{wl}(i, j)} \quad (5)$$

Here, I_{wl} and I'_{wl} denote the grayscale spectral images before and after post-processing, respectively. n and m are the width and height of the image, and k_{avg} is the targeted average intensity, set at 150 for this study. For comparative purposes, spectral upsampling methods involve calculating pixel-wise spectra to generate spectral images, followed by a similar post-processing step for consistency.

2.2.2. Pigment distribution map

In this section, the patch-sized output maps — previously utilized for synthesizing skin images within the training framework — are combined to form comprehensive pigment distribution maps for the entire image. These patch-sized maps are blended with opposite gradient weights applied to adjacent patches for seamless integration. To

evaluate our model's accuracy in pigment analysis, we compare its results with those obtained from the VISIA system. Outcomes from both methods are converted into grayscale images, and their Pearson correlation coefficient [32] is calculated to assess the linear relationship between them.

2.2.3. Pigment simulation

Simulating diverse skin tones and pigment-rich regions involves modifying the pigment distribution maps produced during the reflectance analysis. Adjusting pixel values uniformly in these maps modifies the skin reflectance spectra, which can change the overall skin tone or enhance the contrast in pigment-rich areas.

The ITA method is utilized to quantify variations in skin tone, specifically by averaging the color of the facial region [33] (Fig. 2(B)). ITA calculates the angle between the L^* and a^* axes in the Lab^* color space, assuming a reference L value of 50, and is expressed as:

$$ITA = \arctan\left(\frac{L - 50}{b}\right) \times \frac{180^\circ}{\pi} \quad (6)$$

To quantify pigment levels, we compute the melanin and erythema indices based on the optical principles depicted in Fig. 2(C). Optical density, representing the logarithmic ratio of illuminated to reflected light intensities, indicates the extent of light absorption. The angle of optical density between 620 to 720 nm is proportional to melanin content, while the area under the curve from 560 to 650 nm varies with hemoglobin content [23]. The absorption spectrum of melanin decreases across the visible range, and hemoglobin's absorption drops sharply around 600 to 700 nm. Therefore, optical density around 700 nm is more sensitive to melanin content, and the difference in optical density between 560 and 650 nm is indicative of hemoglobin content. Relating these to RGB wavelengths of the camera sensor [34], melanin and erythema indices from RGB images are computed as follows [23,35]:

$$M.I = OD_R \quad (7)$$

$$E.I = \frac{\sqrt{I(B) \cdot I(R)}}{I(G)} \quad (8)$$

Here, OD_R is the optical density in the red channel, and $[I(R), I(G), I(B)]$ denote the RGB channels of the skin images. The erythema index formula is an improved version over basic methods by reducing the influence of melanin content [35].

2.3. Experiments

The dataset, obtained from the Global Medical Research Center, comprises data from 200 subjects. It includes cross-polarized images captured using the VISIA system (Canfield Scientific Inc.) and pigment maps analyzed by VISIA VAESTRO's RBX-Brown and RBX-Red Processing 1.0 modules [3], designed for detailed pigment analysis. Data acquisition followed national regulations and institutional research ethics policies, with informed consent from all participants. The study received approval from the authors' institution's institutional review board (IRB number : GIRB-21426-PD).

The training conditions were as follows: The dataset was split into a training set (140 subjects) and a test set (60 subjects), with the test set being utilized for the evaluations and analyses presented in the Section 3. Images were cropped and resized to 979×1461 pixels. In the training set, images were divided into 300×300 pixel patches (20 patches per image), allowing extra space for random cropping. For the test set, patches were 256×256 pixels (24 patches per image). For image synthesis, the light source's SPD was set to the standard illuminant 'D65', and the quantum efficiency of the 'Canon 5D Mark II' camera model [36] was used, reflecting the lighting and camera conditions during image acquisition. Data augmentations, implemented using the 'Albumentations' library [37], included horizontal flip, shift, scale, rotation, and random crop. Spectral analysis was performed using

Table 1

Normalized PISV scores from spectral upsampling methods and our model across normal, mild, and noticeable pigment levels.

	Otsu 2018		Jakob 2019		Mallett 2019		Our model	
	Mel	Hb	Mel	Hb	Mel	Hb	Mel	Hb
Normal	1	1	1	1	1	1	1	1
Mild	1.861	1.720	1.594	1.922	1.544	1.497	0.686	1.143
Noticeable	2.233	2.758	1.943	2.520	1.675	1.605	0.626	0.955

the Colour-Science library [38]. The code for this study was developed using the PyCharm integrated development environment (IDE) and written in Python. Computations were done on an RTX 3090 GPU using the PyTorch platform, with training spanning 42 h for 1500 epochs.

For the application frameworks, analysis conditions included: (1) In Spectral Analysis, areas of interest were defined based on melanin and hemoglobin concentrations, classified as normal skin, mild, and noticeable levels. Spectra for these regions were derived using methods from Otsu 2018 [12], Jakob 2019 [13], and Mallett 2019 [14], with full-sized spectral images generated for comprehensive comparison. The inference time of our model was compared with the processing time of spectral upsampling methods. To support spectral interpretation based on pigment absorption properties, we defined our parameter to quantify the curvature of the spectra, with a detailed discussion in the Section 4. Additionally, the wavelength values in result images are provided for interpretive reference, without comparative validation against actual captured images.

(2) For Pigment Distribution Maps, model checkpoints were saved periodically up to 1500 epochs, allowing for an analysis of changes in melanin and hemoglobin distribution maps over time and their correlation with VISIA system outputs. Three subjects with distinct skin tones were selected for comparative analysis against the VISIA system, examining both full-sized images and specific pigment areas.

(3) Pigment Simulation involved adjusting melanin and hemoglobin components to simulate variations in skin tones (by adding -0.8 , -0.2 , $+0.0$, $+0.4$, $+0.8$) and pigment-rich regions (by multiplying by factors of 0.6 , 0.8 , 1.0 , 1.2 , 1.4), thereby altering the reflectance spectrum. These adjustments were quantified through the analysis of curvature variations for skin tones and differences in pigment indices for pigment-rich regions, respectively. For the latter, 30 subjects from the test set were selected, and areas with higher and lower pigmentation were identified. Differences in pigment indices were computed to observe the trends following the multiplication factors, highlighting the model's capability to simulate diverse pigmentary conditions.

3. Results

Reflectance spectra for different pigment levels produced by spectral upsampling methods (Otsu 2018, Jakob 2019, Mallett 2019) and our model are depicted in Fig. 3. Fig. 3(B) illustrates that our model exhibits a reversed 'W' shaped spectrum for normal skin, with changes in the curvature's trend associated with different pigment levels and types. We defined the parameter Pigment-Induced Spectral Variation (PISV) (illustrated in Fig. 3(C)), which quantifies the discrepancy in spectral reflectance at 550 nm and 650 nm wavelengths, reflecting the curvature's flatness. The PISV scores, as determined by the three methods and our model, are normalized relative to their respective values for normal skin, as shown in Table 1. Compared to normal skin, our model showed a reduction in PISV scores to around 0.4 in areas with increased melanin levels. For regions with higher hemoglobin levels, PISV values exhibited slight fluctuations, staying within a 0.15 range. Conversely, the other three methods registered increases in a wide range from 1.6 to 2.7, irrespective of the pigment type.

We also measured the processing times for calculating the reflectance spectrum per pixel on an Intel Xeon Silver 4216 CPU (2.10 GHz). The estimated times for processing 256×256 pixels were

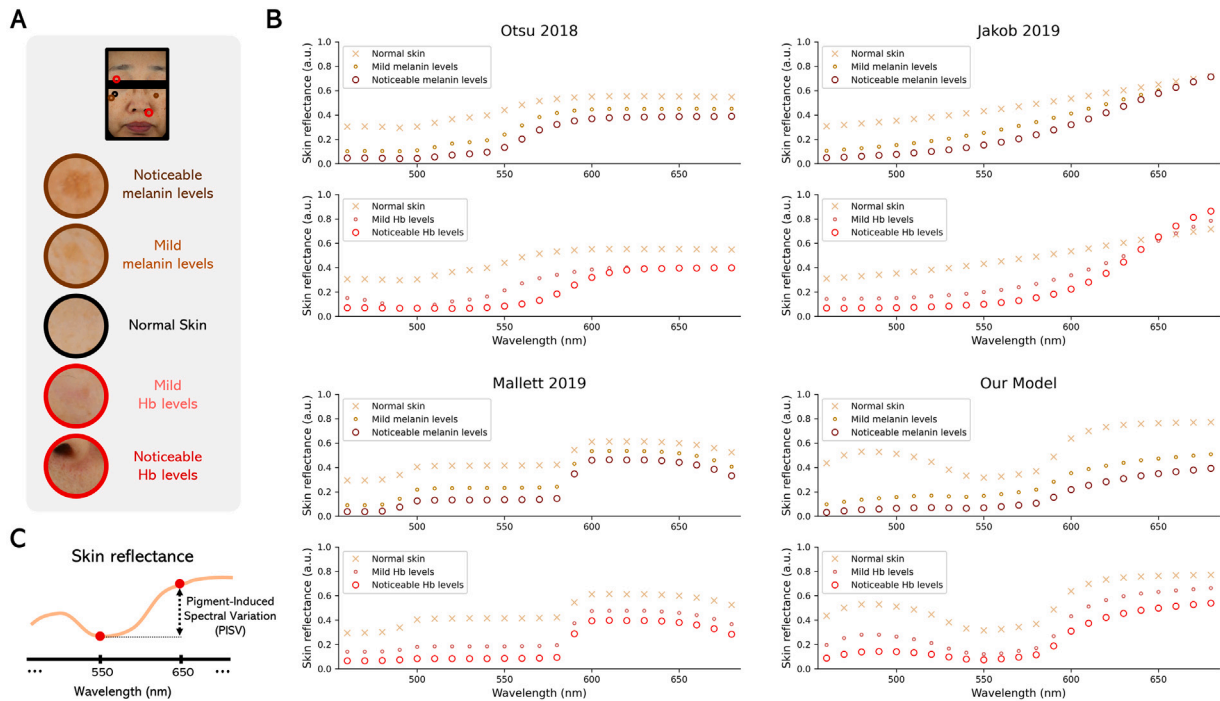


Fig. 3. Comparison of reflectance spectra from our model and spectral upsampling methods. (A) Pigment regions classified as normal skin, mild, and noticeable levels. (B) Reflectance spectra for these pigment regions as generated by three spectral upsampling methods and our model. (C) Illustration defining our parameter for in skin reflectance curvature.

28.9 min for Otsu 2018, 301.4 min for Jakob 2019, and 4.9 min for Mallett 2019. In contrast, our model's inference time on the same CPU was 10.5 s, reducing to 13.6 ms when utilizing an RTX 3090 GPU.

Fig. 4 presents the simulation of spectral images and compares the results from three spectral upsampling methods to those of our model. Fig. 4(A) illustrates the process of generating spectral images from skin reflectance, showcasing both the spectral intensity distributions and the resulting color spectral images produced by our model. Subsequent post-processing enhances the contrast of these spectral images (Fig. 4(B)). We then examined pigment regions classified at 480 nm and 580 nm in the images (Fig. 4(C)). To quantify contrast — indicative of the pigment regions' visibility — we manually selected these areas, calculated their average intensity difference from the background, and normalized the values using average intensity of normal skin as a reference. Overall, the contrast provided by our model exceeded that of the other methods by more than 200-fold, particularly at noticeable Hb levels.

Fig. 5 provides a comparative analysis of pigment distribution maps generated by our model and the VISIA system. Fig. 5(A) displays the correlation coefficients between the two systems across training epochs, determined through post-processing for the test set. At 1300 epochs, the trained model achieved correlation coefficients of 0.913 for melanin and 0.941 for hemoglobin, with a PSNR of 41.76 dB comparing input to synthesized images. Fig. 5(B) illustrates the evolution of detail in pigment maps over the sampled training epochs. Fig. 5(C) offers a comparison of pigment indices derived from conventional RGB images, our model, and the VISIA system across varying skin tones. Average intensity values in the pigment distribution maps were normalized using a baseline ITA of 16.2, highlighting differences in skin tones (ITA 24.6 and 4.2). Both the VISIA system (0.794 and 1.326) and our system (0.891 and 1.265) show a corresponding response in melanin values across different tones, whereas the melanin index from RGB images remains consistent with baseline values in lighter tones (ITA 24.6).

Fig. 6 demonstrates the results of adjusting melanin and hemoglobin components on reflectance spectra, simulating changes in skin tones and pigment-rich regions. These changes were quantified using normalized PISV and pigment indices. To modify overall skin tones, pigment

maps were adjusted by adding factors within the range of $(-0.8, +0.8)$ (Fig. 6(A, B)), leading to changes in normalized PISV: melanin showed tendencies to slightly decrease (0.99 to 0.83), while hemoglobin displayed an increase (0.54 to 1.35) (Fig. 6(C)). To alleviate or enhance pigment-rich regions, pigment maps were modified by multiplication factors within the range of $(\times 0.6, \times 1.4)$ (Fig. 6(D, E)). This adjustment resulted in relatively minor changes in normalized PISV for both melanin (1.08 to 0.92) and hemoglobin (0.96 to 1.00). However, to quantify image contrast in pigment regions, we analyzed the variance in pigment indices between higher and lower pigment regions for 30 subjects within the test set (Fig. 6(F)). A proportional relationship was observed in response to the adjusted factors.

4. Discussion

In Fig. 3(A), we selected pigment regions within the same subject to minimize the effects of skin tone variations across different individuals. Given that our dataset consisted of typical subjects rather than patients with pigmentary disorders, we referred to 'mild' and 'noticeable' levels of pigmentation instead of 'hyper-pigmented'. In Fig. 3(B), it is observed that the extent of skin reflectance curvature increases in areas with higher pigment levels across all three spectral upsampling methods. This observation can be quantified using the PISV score (Table 1), where the score increases by up to 2.8 times from normal skin to noticeable levels, with both melanin and hemoglobin demonstrating this trend. Conversely, skin reflectance spectra in our model tends to flatten in areas with high melanin levels, while maintaining curvature in regions with elevated hemoglobin levels, as also reflected in the PISV scores. Such phenomena can be interpreted from the perspective of pigment absorption spectra.

Fig. 7 conceptually illustrates the absorption spectra of two main pigments: melanin, which gradually decreases, and hemoglobin, which exhibits a 'W' shape with rapidly changing regions. Assuming all other conditions remain constant aside from the absorption by these two main chromophores, we can simplify the analysis of light-tissue interactions. This approach reveals skin reflectance trends that contrast with

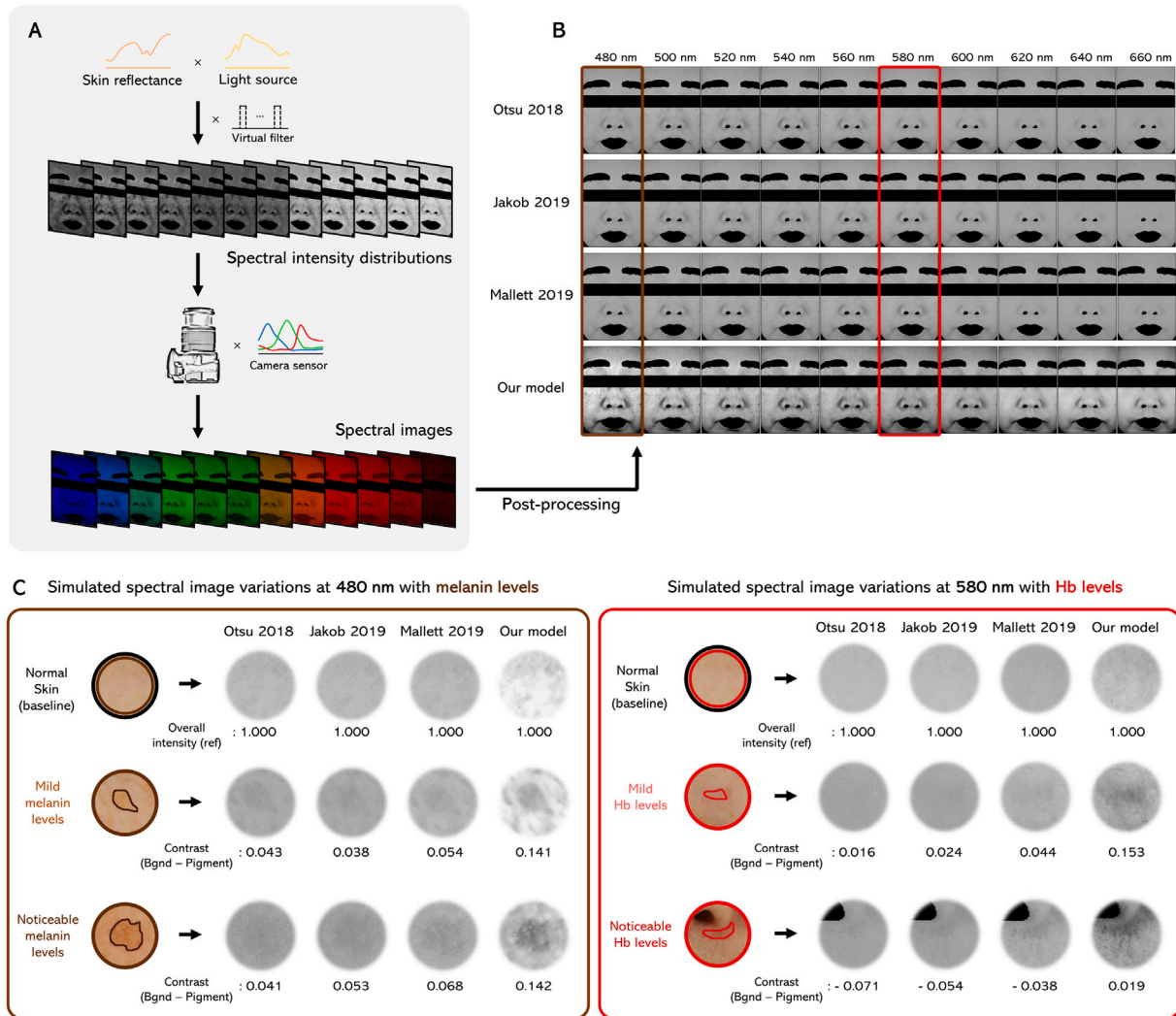


Fig. 4. The generation and comparison of spectral images across pigment regions. (A) Illustration of the spectral image generation from skin reflectance spectra. (B) Spectral images simulated by three different upsampling methods and our model, post-processed for enhanced visibility. (C) Detailed views of pigment regions at 480 nm for melanin and 580 nm for hemoglobin, indicating the intensity contrasts between pigmented areas and the surrounding background.

the absorption spectra of melanin and hemoglobin. Skin reflectance spectra gradually increases in contrast to melanin, creating a reversed 'W' shaped spectrum in opposition to hemoglobin. The ultimate shape of the spectrum varies according to the relative levels of these pigments. As melanin levels increase, the pronounced 'W' shape of hemoglobin becomes flatter and less noticeable, which reduces the PISV score. Conversely, increasing hemoglobin levels enhance the reversed 'W' shape, leading to a higher PISV score. These changes align with findings from previous simulation studies [39]. Although our model's results in Fig. 3(B) show a generally similar trend, the PISV score slightly decreases at noticeable Hb levels. This variation is attributed to a range of factors, including the absorption and scattering by skin components. We manually selected areas with visually distinct pigment levels from the same subject, indicating that reflectance could be differently influenced by various external factors. Ideal outcomes are expected when conditions are uniformly controlled, focusing solely on adjustments in pigment levels, as detailed in the subsequent pigment simulation analysis results.

In Fig. 4, we simulated spectral images from generated skin reflectance spectra and compared them with spectral upsampling methods and our model. Given the practical challenges of modifying expensive commercial equipment to uniformly capture spectral images

and considering our study's focus on skin pigmentation and subsequent analysis of pigment distribution maps, our evaluation aimed to determine how much pigment characteristics are reflected in the simulated spectral images at this stage. We selected wavelengths within bands where melanin and hemoglobin absorption are relatively high and quantified image contrast by calculating the intensity differences between pigment areas and their surrounding background. The contrast in our model exceeded that of other methods by more than 200 times, a result that is qualitatively evident in the images. Particularly at noticeable Hb levels, while other methods yielded negative contrast values, our model maintained a positive contrast of 0.019, although relatively lower than others. This discrepancy could be due to variations in the background skin tone affecting contrast or the contrast in the hemoglobin region being as low as that observed in the background pores. Additionally, the darker appearance of pigment regions at noticeable levels for both pigments possibly results from increased light absorption at higher pigment levels, subsequently reducing the intensity of reflected light.

Fig. 5 presents the comparative analysis results of pigment distribution maps generated by our model and the VISIA system. Ex vivo measurement of pigmentation across numerous subjects' skin areas is

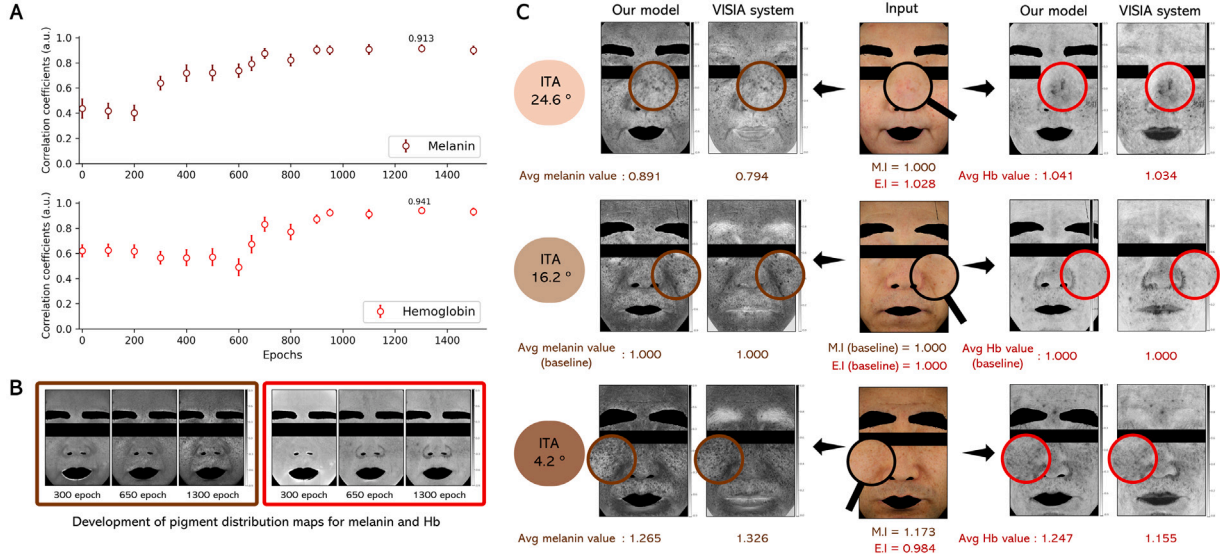


Fig. 5. Comparative analysis of pigment distribution maps from our model and the VISIA system. (A) Trends in correlation coefficients for melanin and hemoglobin across training epochs. (B) Development of melanin and hemoglobin distribution maps over three representative epochs. (C) Pigment distribution maps for melanin and hemoglobin, indicating average intensity values for subjects with varying skin tones.

practically challenging. Hence, the VISIA system, widely recognized in research and diagnostics [19,40], serves as a reference for evaluating our model. In Fig. 5(A), the coefficients for melanin and hemoglobin show varied patterns of stability or fluctuations during the initial training phase, indicating the model's calibration process. This initial instability observed during the early training epochs is likely due to the absence of prepared ground truth data for pigment maps. Strategies such as utilizing pre-trained models could potentially address this issue and enhance the training process's stability. By 1300 epochs, the trained model achieved correlation coefficients of 0.941 for hemoglobin and 0.913 for melanin. The lower value for melanin may necessitate further differentiation efforts due to its shared characteristic of darkness with the facial geometry-induced shadows. The PSNR between input and synthesized images was 41.76 dB, indicating high similarity, as visual differences are generally imperceptible at PSNR values above 30 dB [41]. Fig. 5(B) illustrates the development of images over three sample epochs, with detailed features in the pigment maps becoming increasingly apparent. Fig. 5(C) conducts a comparative analysis of pigment maps for three subjects with different skin tones, utilizing average intensity of pigment maps and conventional RGB image-based pigment indices for quantitative comparison. With decreasing ITA values, both the VISIA system and our model show an increase in melanin value, while hemoglobin's correlation is not significant. In contrast, conventional pigment indices keep the melanin index unchanged from the baseline at ITA 24.6, and the hemoglobin index decreases similarly to ITA values, diverging from the patterns observed with both pigments in the VISIA system. Since ITA classifies skin tone from dark to very light, it is more sensitive to changes induced by melanin than to redness variations related to hemoglobin. This suggests that our model and the VISIA system exhibit trends that correspond more closely to each other than to conventional RGB image-based methods, with the VISIA system serving as the reference in this study.

In Fig. 6, we present the analysis of simulated images depicting variations in skin tones and pigment-rich regions, achieved by numerically adjusting pigment levels. This method, contrasting with the manual selection of visually distinct pigment levels in Fig. 3, uniformly controlled all conditions except for the targeted pigment, allowing for an ideal representation of skin reflectance spectra. Adjusting overall

skin tone involved adding either identical negative or positive values to pigment components. Consequently, melanin (Fig. 6(A)) showed either maintained curvature or flattening, while hemoglobin (Fig. 6(B)) displayed the opposite behavior, becoming more flattened or curved. The PISV scores, normalized to an adjusted factor of 0.0, are detailed in Fig. 6(C). With pigment adjustment factors ranging from -0.8 to $+0.8$, hemoglobin's PISV score varied widely from 0.54 to 1.35, showing an increase relative to the adjusted factor, whereas melanin's score gradually decreased from 0.99 to 0.83. This trend aligns with the theoretical expectations outlined in Fig. 7. Additionally, uniformly multiplying pigment components by identical values to either alleviate or accentuate pigment-rich regions preserved the trend of pigment changes with reduced variations due to minor actual changes in pigment map values. The purpose of this multiplication was to reduce or enhance the contrast between pigment-rich regions and their surrounding backgrounds. Examining the pigment indices differences for these areas across 30 subjects based on adjusted factors (Fig. 6(F)), both melanin and hemoglobin exhibited increased contrast with increasing multiplied values, highlighting more pronounced pigment-rich areas. Centered around the reference value of 1.0, the distribution reflects the diversity of skin tones among the 30 subjects. As multiplied values decreased, melanin's distribution remained consistent, while hemoglobin's reduced, suggesting a move towards lower saturation levels with significantly diminished hemoglobin components.

5. Conclusion

In this study, we validated the effectiveness of our proposed training and application frameworks, focusing on skin pigmentation. This was achieved by analyzing skin reflectance spectra to reflect pigment absorption properties, comparing these spectra with three spectral upsampling methods, assessing pigment distribution maps against those from the reference VISIA system, and creating and analyzing simulations of pigment variation. Our methodology offers several advantages: it provides results comparable to those of specialized clinical equipment without the need for manually prepared ground truth data, demonstrates adaptability by employing images captured by the target device as training data (thus facilitating the development of device-specific

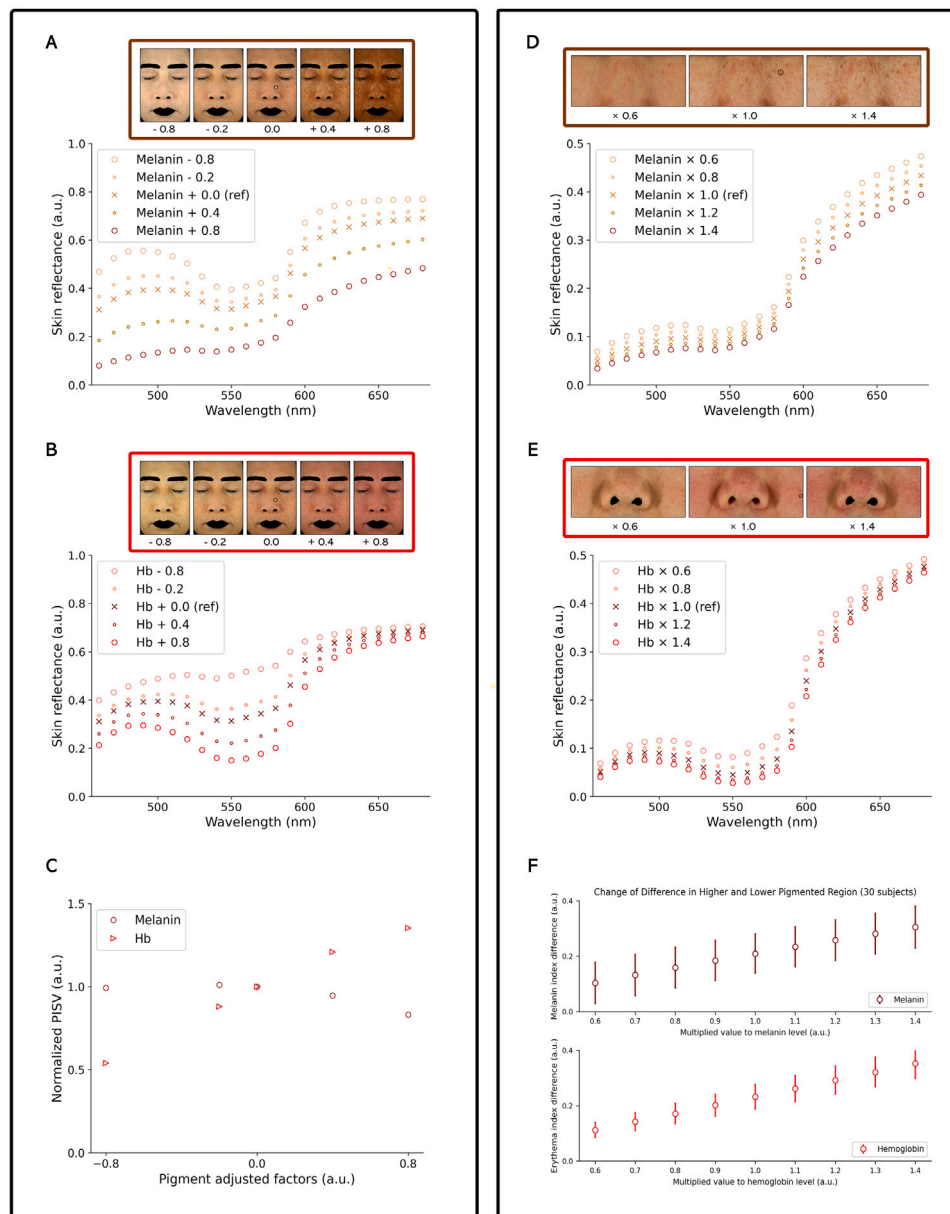


Fig. 6. Illustration of skin tone and pigment concentration variations in response to adjusted factors. (A) Melanin and (B) hemoglobin reflectance spectra changes in the reference region (denoted by black circles) and corresponding overall skin tone images for pigment component adjustments by factors of -0.8 , -0.2 , $+0.0$, $+0.4$, $+0.8$. (C) Normalized PISV relative to pigment adjustment factors. (D) Melanin and (E) hemoglobin reflectance spectra changes in the reference region (indicated by black circles) and pigment-rich region images, varying with pigment component multiplication by factors of 0.6 , 0.8 , 1.0 , 1.2 , 1.4 . (F) Variation in pigment indices for selected areas of higher and lower pigmentation among 30 subjects, corresponding to the multiplication factors applied to melanin and hemoglobin components.

models), and allows numerical control over pigment variations for image simulation. However, this research does not modify expensive clinical equipment to capture actual spectral images, leaving room for performance improvement through such implementation. Additionally, further verification under various image acquisition conditions is necessary to demonstrate the applicability to various devices. This research could potentially be applied in clinical settings in the following ways: automatic detection and quantification of hyperpigmented areas for diagnosis using pigment maps, automation of ground truth generation for hyperpigmentation detection model development, simulation of pigmentation changes and treatment effects, and simulation of skin tone changes after whitening procedures or cosmetic application. To extend the application of this framework to other diseases or fields, future efforts may need to involve modifications to incorporate ground truth elements for enhanced performance and validation, or to capture morphological features of facial diseases more accurately.

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Code and data availability

Due to the involvement of proprietary technology, the code used in this study cannot be publicly released. However, we welcome discussions for research purposes related to the development of the code. Requests for such discussions should be directed to the corresponding author, Semin Kim. The data associated with this article are not publicly available due to privacy and confidentiality constraints. However,

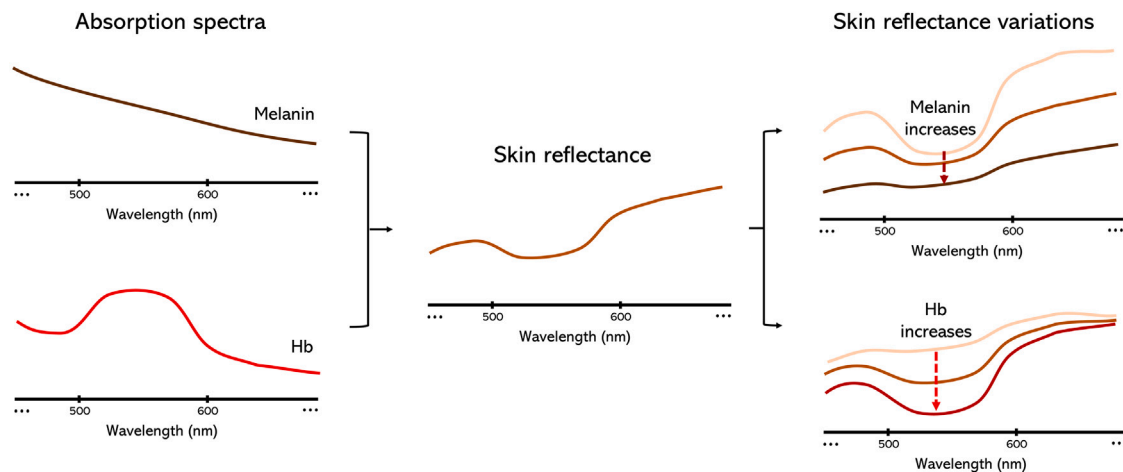


Fig. 7. Conceptual diagram illustrating the variations in skin reflectance spectra associated with different levels of melanin and hemoglobin. As each pigment has a unique absorption spectrum shape, an increase in one pigment's levels results in its characteristic features becoming more pronounced in the skin reflectance spectra.

data can be requested for research purposes through official collaborations with our company. Requests for data access should be directed to the corresponding author, Semin Kim. Prior consent from the subjects and the necessary approvals from the author's institutional review committees will be required.

CRediT authorship contribution statement

Geunho Jung: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Jongha Lee:** Writing – review & editing, Project administration, Data curation. **Semin Kim:** Writing – review & editing, Software, Methodology, Data curation.

Declaration of competing interest

The authors declare that they have no relevant financial interests in this article and no potential conflicts of interest to disclose.

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